

An In-depth Technical Guide to 1-(4-isopropylcyclohexyl)ethanol (CAS: 63767-86-2)

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Compound of Interest

Compound Name: 1-(4-isopropylcyclohexyl)ethanol

CAS No.: 63767-86-2

Cat. No.: B1617246

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For Researchers, Scientists, and Drug Development Professionals

Introduction

1-(4-isopropylcyclohexyl)ethanol, registered under CAS number 63767-86-2, is a synthetic fragrance ingredient valued for its light, floral, and muguet (lily-of-the-valley) scent profile.^{[1][2][3]} This colorless to pale yellow liquid is a stable compound utilized in a variety of applications, including perfumes, cosmetics, and personal care products.^{[2][4][5]} Notably, it exists as four diastereoisomers, with the (-)-1-S-cis-isomer recognized for possessing the most intense and desirable floral aroma.^[1] This technical guide provides a comprehensive overview of its chemical and physical properties, a detailed synthesis protocol, and safety information.

Chemical and Physical Properties

The following table summarizes the key quantitative data for **1-(4-isopropylcyclohexyl)ethanol**.

Property	Value	Source(s)
Molecular Formula	C ₁₁ H ₂₂ O	[4][5][6]
Molecular Weight	170.29 g/mol	[4][5][6]
Appearance	Clear, colorless to pale yellow liquid	[1][2]
Odor	Floral, Lily of the Valley, Rose	[2][5][7]
Boiling Point	Approximately 240 °C	[4]
Density	Estimated at 0.888 g/cm ³	[4]
Specific Gravity	0.9110 - 0.9140	[5]
Flash Point	>100°C	[5]
Refractive Index	1.468 - 1.471	[5]
LogP (Octanol-Water Partition Coefficient)	Estimated at 3.544	[4]
pKa	15.12 ± 0.20 (Predicted)	[6]

Synthesis of 1-(4-isopropylcyclohexyl)ethanol

A common and efficient method for the industrial production of **1-(4-isopropylcyclohexyl)ethanol** is a two-step synthesis starting from the readily available commodity chemical, cumene (isopropylbenzene).[1] This process involves a Friedel-Crafts acylation followed by a hydrogenation reaction.

Experimental Protocol

Step 1: Friedel-Crafts Acylation of Cumene to Cuminone (4-isopropylacetophenone)

This initial step involves the acylation of cumene using either acetyl chloride or acetic anhydride as the acylating agent.[1] The reaction is catalyzed by a Lewis acid or a solid acid catalyst, such as certain zeolites, to promote the electrophilic substitution on the aromatic ring.[1] While traditional Lewis acids like AlCl₃ can be used, solid acid catalysts like zeolites (e.g., ZSM-5, Beta) are often preferred to minimize hazardous waste.[1]

- **Reaction Conditions:** A typical procedure involves reacting cumene with acetic anhydride in a molar ratio of 6:1 (cumene to acetic anhydride).[1] The reaction is carried out in the presence of a zeolite catalyst at a temperature of 150°C.[1] The reaction progress is monitored, and maximal conversion is typically achieved within 1-3 hours.[1]

Step 2: Hydrogenation of Cuminone to **1-(4-isopropylcyclohexyl)ethanol**

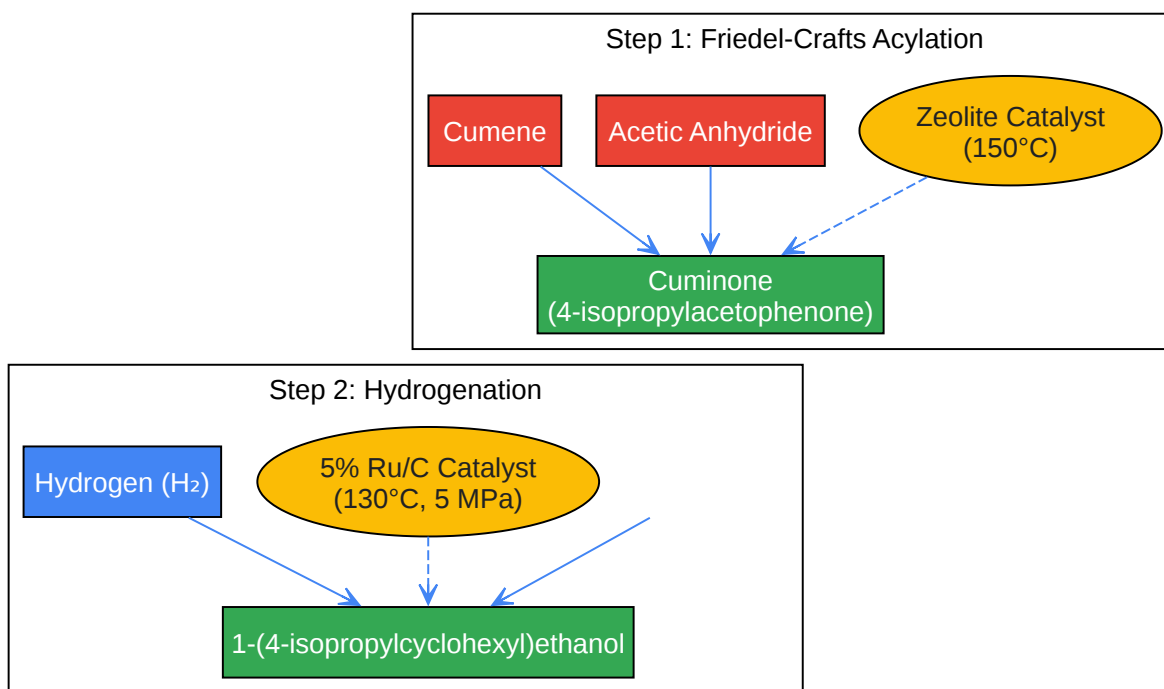
The second step is the hydrogenation of the intermediate, cuminone.[1] This reaction reduces the ketone functional group to a secondary alcohol and saturates the aromatic ring to a cyclohexane ring.

- **Reaction Conditions:** The hydrogenation is performed in a batch autoclave.[1] Cuminone is dissolved in a solvent such as isopropyl alcohol.[1] A ruthenium-based catalyst, typically 5% Ru on a carbon support (Ru/C), is added at a concentration of 5% w/w relative to the cuminone.[1] The reactor is pressurized with hydrogen gas to 5 MPa and heated to 130°C.[1] This process yields **1-(4-isopropylcyclohexyl)ethanol** with high selectivity, producing a mixture of cis and trans isomers.[1] The overall yield of the two-step process can be as high as 75%, with a product purity of 99.5%.[1]

Chemical Structure and Synthesis Workflow

The following diagrams illustrate the chemical structure of **1-(4-isopropylcyclohexyl)ethanol** and the two-step synthesis process.

Caption: Chemical structure of **1-(4-isopropylcyclohexyl)ethanol**.



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Caption: Two-step synthesis of **1-(4-isopropylcyclohexyl)ethanol** from cumene.

Applications

The primary application of **1-(4-isopropylcyclohexyl)ethanol** is in the fragrance industry.[1][4] Its stability and pleasant floral scent make it a versatile ingredient in:

- **Perfumes:** It is a key component in creating lily-of-the-valley and other floral fantasy fragrances.[2][4]
- **Cosmetics and Personal Care Products:** Due to its desirable aroma and stability, it is incorporated into various formulations.[4]
- **Flavoring Agents:** It is also used in food products where floral notes are desired.[4]

Safety and Handling

Based on the Globally Harmonized System (GHS) of Classification and Labelling of Chemicals, **1-(4-isopropylcyclohexyl)ethanol** is classified with the following hazards:

- Skin irritation (Category 2)[6]
- Serious eye irritation (Category 2)[6]
- Hazardous to the aquatic environment, long-term (Chronic) - Category Chronic 2[6]

Precautionary Statements:

- Prevention: Wash hands thoroughly after handling. Wear protective gloves, protective clothing, eye protection, and face protection. Avoid release to the environment.[6]
- Response:
 - If on skin: Wash with plenty of water. If skin irritation occurs, get medical advice/attention. Take off contaminated clothing and wash it before reuse.[6]
 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists, get medical advice/attention.[6]
- Disposal: Collect spillage.[6]

For detailed safety information, it is recommended to consult the full Safety Data Sheet (SDS) from the supplier.

Conclusion

1-(4-isopropylcyclohexyl)ethanol is a significant fragrance ingredient with a well-established synthesis route and a range of applications in the consumer products industry. Its favorable olfactory properties and stability make it a valuable component in many formulations. Understanding its chemical properties, synthesis, and safety is crucial for its effective and responsible use in research and product development.

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